SUBMIND
DATA DICTIONARY

External Entity Description:

EXTERNAL ENTITY DESCRIPTION FORM	
Entry Type: External Entity	
Name: User	
Label: User	**Alias:** End User / Mobile App User/Customer
Input Data Flow: Monthly Reports&Recommendations, Subscription List/Dashboard, Notifications (upcoming payment, trial end, price changes)	
Output Data Flow: Subscription Data, Budget & Notification Preferences, Sign-up / Login Data	
DESCRIPTION	
The User is the external person who interacts with the SubMind system through the mobile or web app. The user provides information such as subscription details, preferences and login data. In return, the system sends back things like subscription dashboards, notifications, AI-based insights, payment confirmations, and verification results. In the SubMind environment (as shown clearly in the Level-0 DFD), the User is the main actor for processes like 2.0 Manage Subscriptions & Preferences, 3.0 Handle Payments & Invoices, 4.0 Generate & Deliver Notifications, and 5.0 Analyze Usage & AI Insight. Every interaction starts with data sent by the user and ends with personalized responses created by the system.	
COMMENT/NOTES	
. User triggers most main processes and provides key input to the system. . Subscription and preference data related to the user are stored in D2 and D3. . All user data is validated before processing.	

Data Flow Description:

DATA FLOW DESCRIPTION FORM	
Entry Type: Data Flow	
Name: Subscription Data	
Label: Subscription Data	**Alias:** Subscription Request / Subscription Update
Origin: User (External Entity)	**Destination:** (2.0) Manage Subscriptions & Preferences

VOLUME AND FREQUENCY

Approximately 10–20 subscription additions, updates, or cancellations occur per user per month. During peak times (e.g., beginning of month), the system may receive hundreds of subscription related requests across all users.

TYPE OF DATA FLOW

✔ Screen	File	Report	Form	Internal

DESCRIPTION

Subscription Data represents the information submitted by the User when adding a new subscription, updating an existing one, or changing renewal preferences.This data flow contains details such as provider name, plan type, pricing, billing cycle, start date, and auto-renew settings. The data is transferred from the User to Process 2.0, where the system validates the input and communicates with Service Providers to verify plan and price information before updating internal data stores.

COMMENT/NOTES

- Includes subscription additions, updates, and cancellations initiated by the user.
- Must be validated before updating the Subscriptions data store (D2).
- This is one of the primary user-to-system flows in the SubM ind application.
- The dashboard and analytics screens are dynamically generated based on the validated subscription data stored in D2

Process Description:

PROCESS DESCRIPTION FORM	
Entry Type: Process	
Name: Update Subscription Data	
Label: Update Subscription Data	
(Process) Number: 2.3	
Input Data Flow: **. Validated Subscription Info** (from Process 2.2 Validate Subscription & Provider Data), **. Fetch Subscription Records** (from D2 Subscriptions)	**Output Data Flow:** **. Update Subscription Records** (to D2 Subscriptions), **Subscription Overview** (to 2.5 Generate Subscription Dashboard) **. Subscription Activity** (to Process 5.0 Analyze Usage & AI Insight) **. Subscription Payment Request** (to 3.0 Handle Payments & Invoices)

DESCRIPTION

This process updates the user's subscription information using validated data received from Process 2.2. Upon receiving Validated Subscription Info, the process retrieves the corresponding subscription record from D2 Subscriptions and updates the fields such as plan, price, renewal settings, and other subscription attributes. After updating the record, the process writes the updated subscription data back to D2.

The updated subscription information is then forwarded as:

- Subscription Overview to Process 2.5 Generate Subscription Dashboard,
- Subscription Activity to Process 5.0 Analyze Usage & AI Insight,
- Subscription Payment Request to Process 3.0 Handle Payments & Invoices when payment-related changes occur (e.g., renewal date, billing cycle, price updates),

ensuring that all downstream modules receive the latest subscription status.

TYPE OF PROCESS

✔ Online Batch Manual

REF. TO NAME:

✔ Structured English Decision Table Decision Tree

BEGIN PROCESS UPDATE_SUBSCRIPTION_DATA

RECEIVE VALIDATED_SUBSCRIPTION_INFO

```
   RETRIEVE SUBSCRIPTION_RECORD FROM DATA_STORE D2

   IF SUBSCRIPTION_RECORD EXISTS THEN
      DO THE FOLLOWING UPDATES:
         - SET plan = VALIDATED_SUBSCRIPTION_INFO.plan
         - SET price = VALIDATED_SUBSCRIPTION_INFO.price
         - SET billing_cycle = VALIDATED_SUBSCRIPTION_INFO.billing_cycle
         - SET renewal_date = VALIDATED_SUBSCRIPTION_INFO.renewal_date
         - SET status = VALIDATED_SUBSCRIPTION_INFO.status
         - UPDATE any additional provider-validated attributes
   ENDIF

   WRITE UPDATED SUBSCRIPTION_RECORD TO DATA_STORE D2

   SEND SUBSCRIPTION_OVERVIEW TO PROCESS 2.5 (GENERATE SUBSCRIPTION
DASHBOARD)

   SEND SUBSCRIPTION_ACTIVITY TO PROCESS 5.0 (ANALYZE USAGE & AI INSIGHT)

   IF UPDATED FIELDS IMPACT PAYMENT THEN
      GENERATE SUBSCRIPTION_PAYMENT_REQUEST
      SEND SUBSCRIPTION_PAYMENT_REQUEST TO PROCESS 3.0 (HANDLE PAYMENTS
& INVOICES)
   ENDIF

END PROCESS
```

COMMENT/NOTES:

- All subscription data entering this process is assumed to be fully validated in Process 2.2.
- If no matching record is found in D2, no update is performed and the system may log the incident.
- Updates are applied atomically to avoid partial or inconsistent records.
- Any change affecting payment (price, billing cycle, renewal date, plan) triggers a Subscription Payment Request to Process 3.0.
- Updated data is forwarded immediately to Process 2.5 and Process 5.0 to ensure real-time dashboard updates and usage analysis.
- Time-sensitive fields must follow consistent formats to ensure compatibility across modules.

Data Store Description:

DATA STORE DESCRIPTION FORM	
Entry Type: Data Store	
Name: Subscriptions	
Label: Subscriptions	**Alias:** Subscriptions / Subscription_Store / Subscription Records

DESCRIPTION:

This data store contains all subscription-related records for every user in the SubMind system. It includes provider name, plan details, price, billing cycle, renewal settings, status, and date fields.
This data store is directly accessed by:
2.3 Update Subscription Data retrieves existing subscription records and writes updated records back into the store.
All subscription creation, modification, cancellation, and renewal updates performed in Process 2.3 rely on this data store.

VOLUME AND FREQUENCY:

- **Read Access:** Approximately 50–120 times daily (records retrieved for update operations in Process 2.3).
- **Write Access:** Approximately 50–120 updates daily (new subscriptions, edits, cancellations).

ATTRIBUTES:

- SubscriptionID
- UserID
- ServiceProvider (e.g., Netflix, Spotify)
- PlanType (Basic, Standard, Premium)
- Price
- RenewalCycle (Monthly / Yearly)
- StartDate
- EndDate
- AutoRenewStatus
- SubscriptionStatus (Active, Paused, Cancelled)
- LastUpdated

COMMENT/NOTES:

- This data store is the main source for all subscription information in the system.
- Only Process 2.3 reads and updates this store. Other processes use the updated outputs but do not access the store directly.
- All updates are made only after the subscription data is successfully validated in Process 2.2.
- The accuracy of this store is important because payment handling, notifications, and usage analysis depend on correct subscription data.
- The “LastUpdated” field is used to track when each subscription record was changed.

Data Structure Description:

DATA STRUCTURE DESCRIPTION FORM

Entry Type: Data Structure

Name: Subscriptions

Label: Subscriptions

Alias: Subscription Records, D2

DESCRIPTION:

Stores detailed information about a user's subscription to a service provider. Includes plan type, provider name, pricing, renewal cycle, dates, status, and metadata used by the system for updates and tracking.
This structure is directly accessed only by Process 2.3 Update Subscription Data to retrieve and modify subscription records.

ATTRIBUTES:

Name	Type	Length	Null
SubscriptionID	Numeric	10	NO
UserID	Numeric	10	NO
ServiceProvider	Alphabetic	30(max)	NO
PlanType	Alphabetic	20(max)	NO
Price	Numeric (Decimal)	6,2	NO
RenewalCycle	Alphabetic	10	NO
StartDate	Date	-	NO
EndDate	Date	-	YES
AutoRenewStatus	Boolean	-	NO
SubscriptionStatus	Alphabetic	10	NO
LastUpdated	Date/Time	-	NO

COMMENT/NOTES:

- SubscriptionID is the primary key for each subscription record.
- EndDate may be empty if the subscription is still active.
- SubscriptionStatus values typically include: Active, Paused, Cancelled.
- AutoRenewStatus is stored as TRUE/FALSE.
- This structure is only accessed directly by Process 2.3. Other processes use the updated data produced after 2.3 completes its operations.
- The LastUpdated field is used to track when the subscription was last modified.

Data Element Description:

DATA ELEMENT DESCRIPTION FORM	
Entry Type: Data Element	
Name: Price	
Label: Price	**Alias:** Subscription Price, Monthly Fee, Plan Cost
Source: Service Providers: (Netflix, Spotify, Amazon…)	**Security:** System-managed
Responsible User: System / SubMind Engine	

DESCRIPTION:

Represents the monthly cost of a subscription plan retrieved from external service providers. This data is used during subscription validation, updates, budgeting calculations, invoice generation, and AI usage/spending insights.

TYPE:

Numeric (Decimal)

TYPE:

✔ Base Derived (Formula:)

VALIDATION CRITERIA:

Continuous	Discrete
. Price ≥ 0.00 . Must contain a valid decimal number (XX.XX) . Accepts two decimal places	. Must match allowed provider plan values . Prices cannot be negative

ATTRIBUTES:

Length:	6 (max) including decimals (e.g., 999.99)
Decimal Point:	2
Input Format:	Numeric decimal (XX.XX), fetched automatically
Output Format:	Local currency format (XX.XX)
Default Value:	None

COMMENT/NOTES:
• Stored in D2 (Subscriptions) as part of each subscription record. • Used directly by Processes 2.2 Validate Subscription & Provider Data and 2.3 Update Subscription Data. • The user cannot manually edit this value; it is updated when the provider changes plan pricing. • Important for budgeting calculations, renewal reminders, invoice generation, and AI spending insights.